



Banks and Gesso, LLC

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September 29, 1999

RECEIVED

OCT 04 1999

Division of Minerals & Geology

Christina L. Kamnikar
Environmental Protection Specialist
Colorado Division of Minerals and Geology
1313 Sherman St., Room 215
Denver, Colorado 80203

Subject: Report on Storage of Cement Kiln Dust at the Southdown, Inc.,
Lyons Plant, Lyons Quarry, Permit # M-77-208

Dear Christina:

Southdown has received letters from you and Harry Posey requesting clarification on certain items contained in the above referenced report. This letter contains Southdown's response (responses are keyed to the numbered items in your September 7, 1999 letter to John Lohr, Southdown, Inc.).

Item No. 1: Sources of Information on Limestone, Shale and Alluvium

The sources of information on the various geologic units present are:

- The NRCS soil survey
- USGS publications and maps on the geology of the area
- Southdown's experience over 30 years of mining the property
- Extensive drilling in the alluvium of St. Vrain Creek by gravel mining interests
- Exploratory drilling at Dowe Flats where the geologic units are the same, and
- Logs of water well drilling in the vicinity

The distances from C-Pit and the sedimentary strata to the alluvium were calculated both from maps and from field measurements. The geomorphic

expression of the alluvium is quite clear compared to the outcrops and subcrops of the sedimentary rocks.

Item No. 2: Modeling of the CKD Disposal Site as an Open vs. Closed Disposal Site.

The model simulations were indeed done under the assumption of a closed disposal site. The rationale for this was that this would be the worst case scenario, the CKD disposal site would be full and the long term, post closure conditions would have to be modeled to determine long term impacts, if any.

Even though Harry Posey has stated that conservative assumptions were used in this analysis, Southdown will commit to interim capping of the disposal site. This management practice is in addition to the other management practices described in the report referenced previously.

The "worst case" leachate volume above which potential off-site water quality impacts could begin to occur is 350,000 cubic feet per year. This volume of water would have to pass through the CKD mass and flow out of the bottom of the disposal site in order to even approach off-site impacts. To prevent this situation from happening, Southdown will place at least 2-feet of compacted shale and clay over the CKD when the disposal area reaches 7 acres in areal extent and 20 feet in thickness.

6.1 million CF

The following section of this letter discusses the results of a additional model simulations that demonstrate that this interim capping practice will never allow the leachate volume to approach the threshold volume of 350,000 cubic feet per year.

Two additional HELP model runs were made to simulate the above described interim capping of the disposed CKD. An initial run was made to simulate an open, partially-filled CKD disposal site. This initial run was simulated as a 7 acre disposal site with approximately 20 feet of CKD capped by an interim 2-foot thick compacted shale/clay cover. Since the disposal site was simulated as an open site, precipitation run-off was not allowed to occur. The default soil type used to simulate the disposed CKD (material texture 10) was the same as that used in the previously submitted HELP model. The compacted shale/clay cover was simulated as a compacted clay (material texture 25) with an effective saturated hydraulic conductivity of $3.60E-06$ centimeters per second (cm/sec). Default evapotranspiration and precipitation data for Denver, Colorado were used in the simulation.

113.5 cm/year = 46.9 in/year

The model results of the open, partially-filled CKD disposal site indicates that the flux of leachate that will be generated at the base of the disposed CKD is about 0.3 inches per year. For this 7 acre disposal site, the annual leachate volume predicted is about 7,592 cubic feet per year. A comparison of the leachate flux generated under the interim capping scenario (0.3 inches per year) with the

leachate flux previously calculated for the closed CKD disposal site (0.74 inches per year), suggests that the interim capping scenario will reduce the leachate flux through the open disposal site by about 40 percent.

An additional HELP model run was made to simulate a closed CKD disposal site that used the interim capping approach. This additional model run was simulated as a 20-acre CKD disposal site with approximately 20 feet of CKD capped by a compacted shale/clay cover overlain by an additional 15 feet of CKD capped by a compacted shale/clay cover. The upper compacted shale/clay cap was covered with two feet of topsoil and vegetated with a poor stand of grass. The topsoil, disposed CKD, and compacted shale/clay layers were simulated using material textures 8, 10, and 25, respectively. Default evapotranspiration and precipitation data for Denver, Colorado were used in the simulation.

The model results of the closed CKD disposal site indicates that the flux of leachate that will be generated at the base of the disposed CKD is about 0.04 inches per year, or an annual leachate volume of 3,232 cubic feet. A comparison of the leachate flux under this modified closure scenario (0.04 inches per year) versus the leachate flux for the previously simulated closed CKD disposal site (0.74 inches per year), suggests that the modified closure scenario will reduce the leachate flux through the closed disposal site by about 95 percent.

The HELP model output for each of the model runs is attached.

Item No. 3: Clarification of HELP Model Data Sets.

The HELP model runs presented in previously submitted Appendix D were made using the HELP model default soil types that most closely approximated the estimated saturated hydraulic conductivities of the topsoil cover, disposed CKD, and the underlying shaley bedrock formations (i.e., Niobrara Formation, Carlile Shale, Greenhorn Limestone, Graneros Shale, and Mowry Shale). The soil type and material textures used to simulate each of these components and the effective saturated hydraulic conductivities simulated are listed in the table below.

Material (Thickness)	Default Soil Type (material texture number)	Effective Saturated Hydraulic Conductivity (cm/sec)
Topsoil Cover (2 feet)	Silt (8)	3.70E-04
Disposed CKD (35 feet)	Clayey Silt (10)	1.20E-04
Shaley Bedrock Formations (700 feet)	Compacted Clay (16)	1.00E-07

Please let me know if this letter adequately responds to your request?

Sincerely,

Paul Banks

Paul Banks, Banks and Gesso, LLC

RICHARD HENRY

Richard Henry, Grant Environmental, Inc.

By PB 9/30/99

PB/kj

Cc: John Lohr, Southdown

Attachments

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**
**      HYDROLOGIC EVALUATION OF LANDFILL PERFORMANCE      **
**      HELP MODEL VERSION 3.07  (1 NOVEMBER 1997)          **
**      DEVELOPED BY ENVIRONMENTAL LABORATORY                **
**      USAE WATERWAYS EXPERIMENT STATION                   **
**      FOR USEPA RISK REDUCTION ENGINEERING LABORATORY      **
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*****
*****
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PRECIPITATION DATA FILE: D:\HELP3\DATA4.D4
TEMPERATURE DATA FILE: D:\HELP3\DATA7.D7
SOLAR RADIATION DATA FILE: D:\HELP3\DATA13.D13
EVAPOTRANSPIRATION DATA: D:\HELP3\DATA11.D11
SOIL AND DESIGN DATA FILE: D:\HELP3\OPEN4.D10
OUTPUT DATA FILE: D:\HELP3\OPEN4.OUT

TIME: 14:17 DATE: 9/26/1999

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*****
TITLE: OPEN, PARTIALLY-FILLED C PIT CKD DISPOSAL SITE, INTERIM CAPPING SCENARIO
*****
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NOTE: INITIAL MOISTURE CONTENT OF THE LAYERS AND SNOW WATER WERE
COMPUTED AS NEARLY STEADY-STATE VALUES BY THE PROGRAM.

LAYER 1
-----TYPE 1 - VERTICAL PERCOLATION LAYER
MATERIAL TEXTURE NUMBER 25

THICKNESS = 24.00 INCHES
 POROSITY = 0.4370 VOL/VOL
 FIELD CAPACITY = 0.3730 VOL/VOL
 WILTING POINT = 0.2660 VOL/VOL
 INITIAL SOIL WATER CONTENT = 0.3267 VOL/VOL
 EFFECTIVE SAT. HYD. COND. = 0.359999990000E-05 CM/SEC

NOTE: SATURATED HYDRAULIC CONDUCTIVITY IS MULTIPLIED BY 1.34
 FOR ROOT CHANNELS IN TOP HALF OF EVAPORATIVE ZONE.

Clay - Silty

LAYER 2
-----TYPE 1 - VERTICAL PERCOLATION LAYER
MATERIAL TEXTURE NUMBER 10

THICKNESS = 240.00 INCHES
 POROSITY = 0.3980 VOL/VOL
 FIELD CAPACITY = 0.2440 VOL/VOL
 WILTING POINT = 0.1360 VOL/VOL
 INITIAL SOIL WATER CONTENT = 0.2440 VOL/VOL
 EFFECTIVE SAT. HYD. COND. = 0.119999997000E-03 CM/SEC

20'
Clayey silty
CK 15

GENERAL DESIGN AND EVAPORATIVE ZONE DATA

NOTE: SCS RUNOFF CURVE NUMBER WAS COMPUTED FROM DEFAULT
 SOIL DATA BASE USING SOIL TEXTURE #25 WITH BARE
 GROUND CONDITIONS, A SURFACE SLOPE OF 2.% AND
 A SLOPE LENGTH OF 300. FEET.

SCS RUNOFF CURVE NUMBER	=	96.80	
FRACTION OF AREA ALLOWING RUNOFF	=	0.0	PERCENT
AREA PROJECTED ON HORIZONTAL PLANE	=	7.000	ACRES
EVAPORATIVE ZONE DEPTH	=	14.0	INCHES
INITIAL WATER IN EVAPORATIVE ZONE	=	4.112	INCHES
UPPER LIMIT OF EVAPORATIVE STORAGE	=	6.118	INCHES
LOWER LIMIT OF EVAPORATIVE STORAGE	=	3.724	INCHES
INITIAL SNOW WATER	=	0.000	INCHES
INITIAL WATER IN LAYER MATERIALS	=	66.401	INCHES
TOTAL INITIAL WATER	=	66.401	INCHES
TOTAL SUBSURFACE INFLOW	=	0.00	INCHES/YEAR

EVAPOTRANSPIRATION AND WEATHER DATA

NOTE: EVAPOTRANSPIRATION DATA WAS OBTAINED FROM
DENVER COLORADO

STATION LATITUDE	=	39.77 DEGREES
MAXIMUM LEAF AREA INDEX	=	0.50
START OF GROWING SEASON (JULIAN DATE)	=	122
END OF GROWING SEASON (JULIAN DATE)	=	286
EVAPORATIVE ZONE DEPTH	=	14.0 INCHES
AVERAGE ANNUAL WIND SPEED	=	8.80 MPH
AVERAGE 1ST QUARTER RELATIVE HUMIDITY	=	54.00 %
AVERAGE 2ND QUARTER RELATIVE HUMIDITY	=	50.00 %
AVERAGE 3RD QUARTER RELATIVE HUMIDITY	=	49.00 %
AVERAGE 4TH QUARTER RELATIVE HUMIDITY	=	54.00 %

NOTE: PRECIPITATION DATA FOR DENVER COLORADO
WAS ENTERED FROM THE DEFAULT DATA FILE.

NOTE: TEMPERATURE DATA WAS SYNTHETICALLY GENERATED USING
COEFFICIENTS FOR DENVER COLORADO

NORMAL MEAN MONTHLY TEMPERATURE (DEGREES FAHRENHEIT)

JAN/JUL	FEB/AUG	MAR/SEP	APR/OCT	MAY/NOV	JUN/DEC
-----	-----	-----	-----	-----	-----
29.50	33.60	38.00	47.40	57.20	67.00
73.30	71.40	62.60	51.90	38.70	32.60

NOTE: SOLAR RADIATION DATA WAS SYNTHETICALLY GENERATED USING
COEFFICIENTS FOR DENVER COLORADO
AND STATION LATITUDE = 39.77 DEGREES

MONTHLY TOTALS (IN INCHES) FOR YEAR 1974

	JAN/JUL	FEB/AUG	MAR/SEP	APR/OCT	MAY/NOV	JUN/DEC
	-----	-----	-----	-----	-----	-----
PRECIPITATION	1.03	0.82	1.32	2.28	0.06	2.01
	2.34	0.16	0.98	1.68	1.06	0.29
RUNOFF	0.000	0.000	0.000	0.000	0.000	0.000
	0.000	0.000	0.000	0.000	0.000	0.000
EVAPOTRANSPIRATION	1.096	0.975	1.086	1.915	0.441	1.688
	1.842	1.168	0.811	1.309	1.086	0.370
PERCOLATION/LEAKAGE THROUGH LAYER 2	0.0000	0.0000	0.0000	0.0000	0.0000	0.0086
	0.0376	0.0934	0.0574	0.0000	0.0000	0.0000

ANNUAL TOTALS FOR YEAR 1974

	INCHES	CU. FEET	PERCENT
PRECIPITATION	14.03	356502.469	100.00
RUNOFF	0.000	0.000	0.00
EVAPOTRANSPIRATION	13.788	350340.781	98.27
PERC./LEAKAGE THROUGH LAYER 2	0.196979	5005.231	1.40
CHANGE IN WATER STORAGE	0.046	1156.392	0.32
SOIL WATER AT START OF YEAR	66.401	1687238.250	
SOIL WATER AT END OF YEAR	66.446	1688394.620	
SNOW WATER AT START OF YEAR	0.000	0.000	0.00
SNOW WATER AT END OF YEAR	0.000	0.000	0.00
ANNUAL WATER BUDGET BALANCE	0.0000	0.067	0.00

MONTHLY TOTALS (IN INCHES) FOR YEAR 1975

	JAN/JUL	FEB/AUG	MAR/SEP	APR/OCT	MAY/NOV	JUN/DEC
PRECIPITATION	0.23 2.78	0.37 2.00	1.19 0.24	1.14 0.30	2.80 1.88	2.11 0.47
RUNOFF	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000
EVAPOTRANSPIRATION	0.298 2.745	0.395 2.770	0.515 0.362	1.473 0.113	1.889 1.226	2.760 0.484
PERCOLATION/LEAKAGE THROUGH LAYER 2	0.0000 0.0000	0.0000 0.0000	0.0000 0.0000	0.0000 0.0000	0.0000 0.0000	0.0259 0.0000

ANNUAL TOTALS FOR YEAR 1975

	INCHES	CU. FEET	PERCENT
PRECIPITATION	15.51	394109.250	100.00
RUNOFF	0.000	0.000	0.00
EVAPOTRANSPIRATION	15.030	381913.375	96.91
PERC./LEAKAGE THROUGH LAYER 2	0.025904	658.220	0.17
CHANGE IN WATER STORAGE	0.454	11537.604	2.93
SOIL WATER AT START OF YEAR	66.446	1688394.620	

Open4

SOIL WATER AT END OF YEAR	66.689	1694573.370	
SNOW WATER AT START OF YEAR	0.000	0.000	0.00
SNOW WATER AT END OF YEAR	0.211	5358.806	1.36
ANNUAL WATER BUDGET BALANCE	0.0000	0.067	0.00

MONTHLY TOTALS (IN INCHES) FOR YEAR 1976

	JAN/JUL	FEB/AUG	MAR/SEP	APR/OCT	MAY/NOV	JUN/DEC
	-----	-----	-----	-----	-----	-----
PRECIPITATION	0.19 2.31	0.54 2.50	1.34 1.88	1.27 0.93	1.34 0.32	0.63 0.16
RUNOFF	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000
EVAPOTRANSPIRATION	0.543 2.482	0.641 2.616	0.895 1.175	1.197 1.177	1.779 0.603	0.579 0.143
PERCOLATION/LEAKAGE THROUGH LAYER 2	0.0000 0.0000	0.0000 0.0382	0.0000 0.0096	0.0000 0.1179	0.0000 0.0601	0.0000 0.0000

ANNUAL TOTALS FOR YEAR 1976

	INCHES	CU. FEET	PERCENT
PRECIPITATION	13.41	340748.250	100.00
RUNOFF	0.000	0.000	0.00
EVAPOTRANSPIRATION	13.832	351475.531	103.15
PERC./LEAKAGE THROUGH LAYER 2	0.225736	5735.954	1.68
CHANGE IN WATER STORAGE	-0.648	-16463.273	-4.83
SOIL WATER AT START OF YEAR	66.689	1694573.370	
SOIL WATER AT END OF YEAR	66.252	1683468.870	
SNOW WATER AT START OF YEAR	0.211	5358.806	1.57
SNOW WATER AT END OF YEAR	0.000	0.000	0.00
ANNUAL WATER BUDGET BALANCE	0.0000	0.036	0.00

MONTHLY TOTALS (IN INCHES) FOR YEAR 1977

JAN/JUL FEB/AUG MAR/SEP APR/OCT MAY/NOV JUN/DEC

PRECIPITATION	0.16	0.27	1.24	2.13	0.34	1.02
	2.98	0.93	0.17	0.09	0.95	0.06
RUNOFF	0.000	0.000	0.000	0.000	0.000	0.000
	0.000	0.000	0.000	0.000	0.000	0.000
EVAPOTRANSPIRATION	0.204	0.248	0.899	2.231	0.569	0.759
	2.781	1.317	0.364	0.161	0.483	0.146
PERCOLATION/LEAKAGE THROUGH LAYER 2	0.0000	0.0000	0.0000	0.0000	0.0000	0.0062
	0.0000	0.0118	0.0000	0.0000	0.0000	0.0000

ANNUAL TOTALS FOR YEAR 1977

	INCHES	CU. FEET	PERCENT
	-----	-----	-----
PRECIPITATION	10.34	262739.469	100.00
RUNOFF	0.000	0.000	0.00
EVAPOTRANSPIRATION	10.163	258232.969	98.28
PERC./LEAKAGE THROUGH LAYER 2	0.017961	456.391	0.17
CHANGE IN WATER STORAGE	0.159	4050.184	1.54
SOIL WATER AT START OF YEAR	66.252	1683468.870	
SOIL WATER AT END OF YEAR	66.412	1687519.120	
SNOW WATER AT START OF YEAR	0.000	0.000	0.00

SNOW WATER AT END OF YEAR	0.000	0.000	0.00
ANNUAL WATER BUDGET BALANCE	0.0000	-0.062	0.00

MONTHLY TOTALS (IN INCHES) FOR YEAR 1978

	JAN/JUL	FEB/AUG	MAR/SEP	APR/OCT	MAY/NOV	JUN/DEC
	-----	-----	-----	-----	-----	-----
PRECIPITATION	0.27	0.27	1.07	1.42	3.86	1.17
	0.54	0.26	0.07	1.45	0.50	0.79
RUNOFF	0.000	0.000	0.000	0.000	0.000	0.000
	0.000	0.000	0.000	0.000	0.000	0.000
EVAPOTRANSPIRATION	0.286	0.313	0.902	1.428	2.119	1.550
	0.799	0.812	0.072	0.466	0.990	1.057
PERCOLATION/LEAKAGE THROUGH	0.0000	0.0000	0.0000	0.0000	0.0010	0.0125
LAYER 2	0.0538	0.1887	0.1753	0.2053	0.1923	0.1985

ANNUAL TOTALS FOR YEAR 1978

INCHES

CU. FEET

PERCENT

Open4

PRECIPITATION	11.67	296534.781	100.00
RUNOFF	0.000	0.000	0.00
EVAPOTRANSPIRATION	10.793	274253.250	92.49
PERC./LEAKAGE THROUGH LAYER 2	1.027332	26104.518	8.80
CHANGE IN WATER STORAGE	-0.150	-3823.171	-1.29
SOIL WATER AT START OF YEAR	66.412	1687519.120	
SOIL WATER AT END OF YEAR	66.261	1683695.870	
SNOW WATER AT START OF YEAR	0.000	0.000	0.00
SNOW WATER AT END OF YEAR	0.000	0.000	0.00
ANNUAL WATER BUDGET BALANCE	0.0000	0.167	0.00

AVERAGE MONTHLY VALUES IN INCHES FOR YEARS 1974 THROUGH 1978

	JAN/JUL	FEB/AUG	MAR/SEP	APR/OCT	MAY/NOV	JUN/DEC
PRECIPITATION						
TOTALS	0.38	0.45	1.23	1.65	1.68	1.39
	2.19	1.17	0.67	0.89	0.94	0.35

STD. DEVIATIONS	0.37	0.23	0.11	0.52	1.62	0.65
	0.97	1.04	0.77	0.69	0.61	0.29
RUNOFF						

TOTALS	0.000	0.000	0.000	0.000	0.000	0.000
	0.000	0.000	0.000	0.000	0.000	0.000
STD. DEVIATIONS	0.000	0.000	0.000	0.000	0.000	0.000
	0.000	0.000	0.000	0.000	0.000	0.000
EVAPOTRANSPIRATION						

TOTALS	0.486	0.514	0.859	1.649	1.359	1.467
	2.130	1.737	0.557	0.645	0.878	0.440
STD. DEVIATIONS	0.364	0.298	0.209	0.416	0.791	0.868
	0.834	0.894	0.435	0.564	0.319	0.375
PERCOLATION/LEAKAGE THROUGH LAYER 2						

TOTALS	0.0000	0.0000	0.0000	0.0000	0.0002	0.0106
	0.0183	0.0664	0.0485	0.0646	0.0505	0.0397
STD. DEVIATIONS	0.0000	0.0000	0.0000	0.0000	0.0004	0.0097
	0.0257	0.0773	0.0748	0.0937	0.0834	0.0888

AVERAGE ANNUAL TOTALS & (STD. DEVIATIONS) FOR YEARS 1974 THROUGH 1978

	INCHES	CU. FEET	PERCENT
	-----	-----	-----

Open4

PRECIPITATION	12.99	(2.023)	330126.9	100.00
RUNOFF	0.000	(0.0000)	0.00	0.000
EVAPOTRANSPIRATION	12.721	(2.1193)	323243.19	97.915
PERCOLATION/LEAKAGE THROUGH LAYER 2	0.29878	(0.41827)	7592.062	2.29974
CHANGE IN WATER STORAGE	-0.028	(0.4099)	-708.45	-0.215

PEAK DAILY VALUES FOR YEARS 1974 THROUGH 1978

	(INCHES)	(CU. FT.)
PRECIPITATION	1.79	45483.898
RUNOFF	0.000	0.0000
PERCOLATION/LEAKAGE THROUGH LAYER 2	0.011504	292.32217
SNOW WATER	1.19	30343.0996
MAXIMUM VEG. SOIL WATER (VOL/VOL)		0.4276
MINIMUM VEG. SOIL WATER (VOL/VOL)		0.2660

FINAL WATER STORAGE AT END OF YEAR 1978

LAYER	(INCHES)	(VOL/VOL)
1	7.7026	0.3209
2	58.5586	0.2440
SNOW WATER	0.000	

Closed1

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HYDROLOGIC EVALUATION OF LANDFILL PERFORMANCE
HELP MODEL VERSION 3.07 (1 NOVEMBER 1997)
DEVELOPED BY ENVIRONMENTAL LABORATORY
USAE WATERWAYS EXPERIMENT STATION
FOR USEPA RISK REDUCTION ENGINEERING LABORATORY
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PRECIPITATION DATA FILE: D:\HELP3\DATA4.D4
TEMPERATURE DATA FILE: D:\HELP3\DATA7.D7
SOLAR RADIATION DATA FILE: D:\HELP3\DATA13.D13
EVAPOTRANSPIRATION DATA: D:\HELP3\DATA11.D11
SOIL AND DESIGN DATA FILE: D:\HELP3\CLOSED1.D10
OUTPUT DATA FILE: D:\HELP3\CLOSED1.OUT

TIME: 10:16 DATE: 9/26/1999

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TITLE: CLOSED C PIT CKD DISPOSAL SITE, MODIFIED CLOSURE SCENARIO

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NOTE: INITIAL MOISTURE CONTENT OF THE LAYERS AND SNOW WATER WERE
COMPUTED AS NEARLY STEADY-STATE VALUES BY THE PROGRAM.

LAYER 1

TYPE 1 - VERTICAL PERCOLATION LAYER

MATERIAL TEXTURE NUMBER 8

THICKNESS = 24.00 INCHES
 POROSITY = 0.4630 VOL/VOL
 FIELD CAPACITY = 0.2320 VOL/VOL
 WILTING POINT = 0.1160 VOL/VOL
 INITIAL SOIL WATER CONTENT = 0.1752 VOL/VOL
 EFFECTIVE SAT. HYD. COND. = 0.369999994000E-03 CM/SEC

NOTE: SATURATED HYDRAULIC CONDUCTIVITY IS MULTIPLIED BY 1.34
 FOR ROOT CHANNELS IN TOP HALF OF EVAPORATIVE ZONE.

2'
Topsoil

LAYER 2

TYPE 1 - VERTICAL PERCOLATION LAYER

MATERIAL TEXTURE NUMBER 25

THICKNESS = 24.00 INCHES
 POROSITY = 0.4370 VOL/VOL
 FIELD CAPACITY = 0.3730 VOL/VOL
 WILTING POINT = 0.2660 VOL/VOL
 INITIAL SOIL WATER CONTENT = 0.3730 VOL/VOL
 EFFECTIVE SAT. HYD. COND. = 0.359999990000E-05 CM/SEC

2'
Clay-shale

LAYER 3

TYPE 1 - VERTICAL PERCOLATION LAYER

MATERIAL TEXTURE NUMBER 10

THICKNESS = 210.00 INCHES

16' CKD

Closed1

POROSITY	=	0.3980	VOL/VOL
FIELD CAPACITY	=	0.2440	VOL/VOL
WILTING POINT	=	0.1360	VOL/VOL
INITIAL SOIL WATER CONTENT	=	0.2440	VOL/VOL
EFFECTIVE SAT. HYD. COND.	=	0.119999997000E-03	CM/SEC

LAYER 4

TYPE 1 - VERTICAL PERCOLATION LAYER

MATERIAL TEXTURE NUMBER 25

THICKNESS	=	24.00	INCHES
POROSITY	=	0.4370	VOL/VOL
FIELD CAPACITY	=	0.3730	VOL/VOL
WILTING POINT	=	0.2660	VOL/VOL
INITIAL SOIL WATER CONTENT	=	0.3730	VOL/VOL
EFFECTIVE SAT. HYD. COND.	=	0.359999990000E-05	CM/SEC

any oil

LAYER 5

TYPE 1 - VERTICAL PERCOLATION LAYER

MATERIAL TEXTURE NUMBER 10

THICKNESS	=	210.00	INCHES
POROSITY	=	0.3980	VOL/VOL
FIELD CAPACITY	=	0.2440	VOL/VOL
WILTING POINT	=	0.1360	VOL/VOL
INITIAL SOIL WATER CONTENT	=	0.2440	VOL/VOL
EFFECTIVE SAT. HYD. COND.	=	0.119999997000E-03	CM/SEC

CCP 1/4

GENERAL DESIGN AND EVAPORATIVE ZONE DATA

NOTE: SCS RUNOFF CURVE NUMBER WAS COMPUTED FROM DEFAULT
 SOIL DATA BASE USING SOIL TEXTURE # 8 WITH A
 POOR STAND OF GRASS, A SURFACE SLOPE OF 2. %
 AND A SLOPE LENGTH OF 300. FEET.

SCS RUNOFF CURVE NUMBER	=	85.80	
FRACTION OF AREA ALLOWING RUNOFF	=	100.0	PERCENT
AREA PROJECTED ON HORIZONTAL PLANE	=	20.000	ACRES
EVAPORATIVE ZONE DEPTH	=	14.0	INCHES
INITIAL WATER IN EVAPORATIVE ZONE	=	1.884	INCHES
UPPER LIMIT OF EVAPORATIVE STORAGE	=	6.482	INCHES
LOWER LIMIT OF EVAPORATIVE STORAGE	=	1.624	INCHES
INITIAL SNOW WATER	=	0.000	INCHES
INITIAL WATER IN LAYER MATERIALS	=	124.586	INCHES
TOTAL INITIAL WATER	=	124.586	INCHES
TOTAL SUBSURFACE INFLOW	=	0.00	INCHES/YEAR

EVAPOTRANSPIRATION AND WEATHER DATA

NOTE: EVAPOTRANSPIRATION DATA WAS OBTAINED FROM
 DENVER COLORADO

STATION LATITUDE	=	39.77	DEGREES
MAXIMUM LEAF AREA INDEX	=	0.50	
START OF GROWING SEASON (JULIAN DATE)	=	122	
END OF GROWING SEASON (JULIAN DATE)	=	286	
EVAPORATIVE ZONE DEPTH	=	14.0	INCHES
AVERAGE ANNUAL WIND SPEED	=	8.80	MPH
AVERAGE 1ST QUARTER RELATIVE HUMIDITY	=	54.00	%
AVERAGE 2ND QUARTER RELATIVE HUMIDITY	=	50.00	%
AVERAGE 3RD QUARTER RELATIVE HUMIDITY	=	49.00	%
AVERAGE 4TH QUARTER RELATIVE HUMIDITY	=	54.00	%

NOTE: PRECIPITATION DATA FOR DENVER COLORADO
WAS ENTERED FROM THE DEFAULT DATA FILE.

NOTE: TEMPERATURE DATA WAS SYNTHETICALLY GENERATED USING
COEFFICIENTS FOR DENVER COLORADO

NORMAL MEAN MONTHLY TEMPERATURE (DEGREES FAHRENHEIT)

JAN/JUL	FEB/AUG	MAR/SEP	APR/OCT	MAY/NOV	JUN/DEC
29.50	33.60	38.00	47.40	57.20	67.00
73.30	71.40	62.60	51.90	38.70	32.60

NOTE: SOLAR RADIATION DATA WAS SYNTHETICALLY GENERATED USING
COEFFICIENTS FOR DENVER COLORADO
AND STATION LATITUDE = 39.77 DEGREES

MONTHLY TOTALS (IN INCHES) FOR YEAR 1974

	JAN/JUL	FEB/AUG	MAR/SEP	APR/OCT	MAY/NOV	JUN/DEC
PRECIPITATION	1.03 2.34	0.82 0.16	1.32 0.98	2.28 1.68	0.06 1.06	2.01 0.29
RUNOFF	0.233 0.048	0.000 0.000	0.000 0.000	0.001 0.000	0.000 0.000	0.179 0.000
EVAPOTRANSPIRATION	0.470	0.917	0.953	2.483	0.606	1.938

Closed1

	1.328	1.125	0.417	1.637	1.340	0.467
PERCOLATION/LEAKAGE THROUGH	0.0000	0.0000	0.0000	0.0000	0.0000	0.0016
LAYER 5	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

ANNUAL TOTALS FOR YEAR 1974

	INCHES	CU. FEET	PERCENT
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PRECIPITATION	14.03	1018578.440	100.00
RUNOFF	0.461	33475.859	3.29
EVAPOTRANSPIRATION	13.680	993144.000	97.50
PERC./LEAKAGE THROUGH LAYER 5	0.001591	115.519	0.01
CHANGE IN WATER STORAGE	-0.112	-8157.751	-0.80
SOIL WATER AT START OF YEAR	124.586	9044934.000	
SOIL WATER AT END OF YEAR	124.473	9036776.000	
SNOW WATER AT START OF YEAR	0.000	0.000	0.00
SNOW WATER AT END OF YEAR	0.000	0.000	0.00
ANNUAL WATER BUDGET BALANCE	0.0000	0.839	0.00

MONTHLY TOTALS (IN INCHES) FOR YEAR 1975

	JAN/JUL	FEB/AUG	MAR/SEP	APR/OCT	MAY/NOV	JUN/DEC
PRECIPITATION	0.23 2.78	0.37 2.00	1.19 0.24	1.14 0.30	2.80 1.88	2.11 0.47
RUNOFF	0.000 0.000	0.000 0.059	0.000 0.000	0.000 0.000	0.024 0.000	0.000 0.057
EVAPOTRANSPIRATION	0.335 2.557	0.393 2.778	0.330 0.228	1.404 0.129	1.719 1.135	3.119 0.399
PERCOLATION/LEAKAGE THROUGH LAYER 5	0.0000 0.0014	0.0000 0.0013	0.0000 0.0000	0.0000 0.0000	0.0000 0.0000	0.0000 0.0000

ANNUAL TOTALS FOR YEAR 1975

	INCHES	CU. FEET	PERCENT
PRECIPITATION	15.51	1126026.370	100.00
RUNOFF	0.141	10220.204	0.91
EVAPOTRANSPIRATION	14.527	1054672.120	93.66

Closed1

PERC./LEAKAGE THROUGH LAYER 5	0.002744	199.195	0.02
CHANGE IN WATER STORAGE	0.839	60934.437	5.41
SOIL WATER AT START OF YEAR	124.473	9036776.000	
SOIL WATER AT END OF YEAR	125.313	9097710.000	
SNOW WATER AT START OF YEAR	0.000	0.000	0.00
SNOW WATER AT END OF YEAR	0.000	0.000	0.00
ANNUAL WATER BUDGET BALANCE	0.0000	0.525	0.00

MONTHLY TOTALS (IN INCHES) FOR YEAR 1976

	JAN/JUL	FEB/AUG	MAR/SEP	APR/OCT	MAY/NOV	JUN/DEC
PRECIPITATION	0.19 2.31	0.54 2.50	1.34 1.88	1.27 0.93	1.34 0.32	0.63 0.16
RUNOFF	0.000 0.084	0.000 0.150	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000
EVAPOTRANSPIRATION	0.288 2.027	0.611 2.912	0.913 0.987	1.144 1.065	2.048 0.466	0.887 0.162
PERCOLATION/LEAKAGE THROUGH LAYER 5	0.0000 0.0000	0.0000 0.0000	0.0000 0.0000	0.0000 0.0000	0.0000 0.0000	0.0000 0.0000

Closed1

ANNUAL TOTALS FOR YEAR 1976

	INCHES	CU. FEET	PERCENT
PRECIPITATION	13.41	973566.375	100.00
RUNOFF	0.234	16975.234	1.74
EVAPOTRANSPIRATION	13.510	980798.187	100.74
PERC./LEAKAGE THROUGH LAYER 5	0.000000	0.000	0.00
CHANGE IN WATER STORAGE	-0.333	-24207.385	-2.49
SOIL WATER AT START OF YEAR	125.313	9097710.000	
SOIL WATER AT END OF YEAR	124.979	9073503.000	
SNOW WATER AT START OF YEAR	0.000	0.000	0.00
SNOW WATER AT END OF YEAR	0.000	0.000	0.00
ANNUAL WATER BUDGET BALANCE	0.0000	0.372	0.00

MONTHLY TOTALS (IN INCHES) FOR YEAR 1977

	JAN/JUL	FEB/AUG	MAR/SEP	APR/OCT	MAY/NOV	JUN/DEC
PRECIPITATION	0.16 2.98	0.27 0.93	1.24 0.17	2.13 0.09	0.34 0.95	1.02 0.06
RUNOFF	0.000 0.003	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000	0.000 0.000
EVAPOTRANSPIRATION	0.230 3.022	0.328 1.208	1.024 0.450	2.238 0.087	0.784 0.440	0.590 0.185
PERCOLATION/LEAKAGE THROUGH LAYER 5	0.0000 0.0000	0.0000 0.0000	0.0000 0.0000	0.0000 0.0000	0.0000 0.0000	0.0112 0.0000

ANNUAL TOTALS FOR YEAR 1977

	INCHES	CU. FEET	PERCENT
PRECIPITATION	10.34	750684.250	100.00
RUNOFF	0.003	215.035	0.03
EVAPOTRANSPIRATION	10.586	768516.500	102.38
PERC./LEAKAGE THROUGH LAYER 5	0.011178	811.519	0.11
CHANGE IN WATER STORAGE	-0.260	-18858.984	-2.51

Closed1

SOIL WATER AT START OF YEAR	124.979	9073503.000	
SOIL WATER AT END OF YEAR	124.720	9054644.000	
SNOW WATER AT START OF YEAR	0.000	0.000	0.00
SNOW WATER AT END OF YEAR	0.000	0.000	0.00
ANNUAL WATER BUDGET BALANCE	0.0000	0.158	0.00

MONTHLY TOTALS (IN INCHES) FOR YEAR 1978

	JAN/JUL	FEB/AUG	MAR/SEP	APR/OCT	MAY/NOV	JUN/DEC
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PRECIPITATION	0.27 0.54	0.27 0.26	1.07 0.07	1.42 1.45	3.86 0.50	1.17 0.79
RUNOFF	0.000 0.000	0.000 0.000	0.000 0.000	0.004 0.000	0.069 0.000	0.000 0.000
EVAPOTRANSPIRATION	0.295 0.361	0.345 0.528	0.607 0.071	1.435 0.254	2.804 0.301	1.889 1.232
PERCOLATION/LEAKAGE THROUGH LAYER 5	0.0000 0.0000	0.0000 0.0000	0.0000 0.0313	0.0000 0.0450	0.0000 0.0545	0.0076 0.0687

ANNUAL TOTALS FOR YEAR 1978

	INCHES	CU. FEET	PERCENT
PRECIPITATION	11.67	847242.187	100.00
RUNOFF	0.073	5282.355	0.62
EVAPOTRANSPIRATION	10.121	734814.812	86.73
PERC./LEAKAGE THROUGH LAYER 5	0.207066	15032.994	1.77
CHANGE IN WATER STORAGE	1.269	92111.469	10.87
SOIL WATER AT START OF YEAR	124.720	9054644.000	
SOIL WATER AT END OF YEAR	125.988	9146755.000	
SNOW WATER AT START OF YEAR	0.000	0.000	0.00
SNOW WATER AT END OF YEAR	0.000	0.000	0.00
ANNUAL WATER BUDGET BALANCE	0.0000	0.584	0.00

AVERAGE MONTHLY VALUES IN INCHES FOR YEARS 1974 THROUGH 1978

JAN/JUL	FEB/AUG	MAR/SEP	APR/OCT	MAY/NOV	JUN/DEC
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PRECIPITATION

TOTALS	0.38	0.45	1.23	1.65	1.68	1.39
	2.19	1.17	0.67	0.89	0.94	0.35
STD. DEVIATIONS	0.37	0.23	0.11	0.52	1.62	0.65
	0.97	1.04	0.77	0.69	0.61	0.29

RUNOFF

TOTALS	0.047	0.000	0.000	0.001	0.019	0.036
	0.027	0.042	0.000	0.000	0.000	0.011
STD. DEVIATIONS	0.104	0.000	0.000	0.002	0.030	0.080
	0.038	0.066	0.000	0.000	0.000	0.026

EVAPOTRANSPIRATION

TOTALS	0.324	0.519	0.765	1.741	1.592	1.685
	1.859	1.710	0.431	0.634	0.736	0.489
STD. DEVIATIONS	0.090	0.249	0.291	0.583	0.911	1.000
	1.048	1.070	0.347	0.687	0.467	0.436

PERCOLATION/LEAKAGE THROUGH LAYER 5

TOTALS	0.0000	0.0000	0.0000	0.0000	0.0000	0.0041
	0.0003	0.0003	0.0063	0.0090	0.0109	0.0137
STD. DEVIATIONS	0.0000	0.0000	0.0000	0.0000	0.0000	0.0051
	0.0006	0.0006	0.0140	0.0201	0.0244	0.0307

AVERAGE ANNUAL TOTALS & (STD. DEVIATIONS) FOR YEARS 1974 THROUGH 1978

	INCHES		CU. FEET	PERCENT
PRECIPITATION	12.99	(2.023)	943219.7	100.00
RUNOFF	0.182	(0.1777)	13233.74	1.403
EVAPOTRANSPIRATION	12.485	(1.9901)	906389.12	96.095
PERCOLATION/LEAKAGE THROUGH LAYER 5	0.04452	(0.09097)	3231.845	0.34264
CHANGE IN WATER STORAGE	0.281	(0.7266)	20364.36	2.159

PEAK DAILY VALUES FOR YEARS 1974 THROUGH 1978

	(INCHES)	(CU. FT.)
PRECIPITATION	1.79	129954.000
RUNOFF	0.179	12994.3125
PERCOLATION/LEAKAGE THROUGH LAYER 5	0.003314	240.56551
SNOW WATER	1.19	86694.5703
MAXIMUM VEG. SOIL WATER (VOL/VOL)		0.3108

Closed1

MINIMUM VEG. SOIL WATER (VOL/VOL)

0.1160

FINAL WATER STORAGE AT END OF YEAR 1978

LAYER	(INCHES)	(VOL/VOL)
1	4.9179	0.2049
2	8.9520	0.3730
3	51.4778	0.2451
4	9.3045	0.3877
5	51.3362	0.2445
SNOW WATER	0.000	
